u_Pointwise_Conv

Pointwise_Conv

u_Ctrl_FSM

Ctrl_FSM

clk

done

en

first_ch_tile
last_ch_tile

load_mode

rst

start

calc_cyc_cnt[15:0]

ch_tile_cnt[15:0]

filter_tile_cnt[15:0]

load_cnt[15:0]

valid[7:0]

c_state_reg[2:0]

RTL_REG_ASYNC C

CLR D Q

calc_cyc_cnt_i

RTL_MUX S

S=1'b1 I0[15:0] S=default I1[15:0] O[15:0]

calc_cyc_cnt_reg[15:0]

RTL_REG_ASYNC C

CLR D Q +

calc_cyc_cnt0_i

RTL_ADD

I1

I0[15:0] O[15:0]

calc_cyc_cnt1_i RTL_OR

I0

I1 O =

calc_cyc_cnt2_i

RTL_EQO

I0[2:0] V=B"010" I1[2:0] =

calc_cyc_cnt2_i__0

RTL_EQO

I0[2:0] V=B"011" I1[2:0]

ch_tile_cnt_i

RTL_MUX S

S=1'b1 I0[15:0] S=default I1[15:0] O[15:0]

ch_tile_cnt_i__0

RTL_MUX

S=3'b000 I0[15:0] S=default I1[15:0] O[15:0] S[2:0]

ch_tile_cnt_i__1

RTL_MUX

S=3'b000 I0

S=default I1 O

S[2:0]

ch_tile_cnt_reg[15:0]

RTL_REG_ASYNC C

CE

CLR D Q +

ch_tile_cnt0_i

RTL_ADD

I1

I0[15:0] O[15:0]

ch_tile_cnt0_i__0 RTL_AND

I0

I1 O =

ch_tile_cnt1_i

RTL_EQO

I0[15:0] V=B"0000000011011010" I1[15:0]

done_reg

RTL_REG_ASYNC C

CLR D Q

en_reg

RTL_REG_ASYNC C

CLR D Q

en0_i RTL_OR

I0

I1 O

filter_tile_cnt_i

RTL_MUX

S=3'b000 I0[15:0] S=default I1[15:0] O[15:0] S[2:0]

filter_tile_cnt_i__0

RTL_MUX

S=1'b1 I0

S=default I1 O S

filter_tile_cnt_i__1

RTL_MUX

S=1'b1 I0

S=default I1 O S

filter_tile_cnt_i__2

RTL_MUX

S=3'b000 I0

S=default I1 O

S[2:0]

filter_tile_cnt_reg[15:0]

RTL_REG_ASYNC C

CE

CLR D Q +

filter_tile_cnt0_i

RTL_ADD

I1

I0[15:0] O[15:0] =

first_ch_tile_i

RTL_EQO

I0[15:0]

I1[15:0] =

last_ch_tile_i

RTL_EQO

I0[15:0] V=B"0000000000000111" I1[15:0]

load_cnt_i

RTL_MUX

S=3'b001 I0[15:0] S=default I1[15:0] O[15:0] S[2:0]

load_cnt_reg[15:0]

RTL_REG_ASYNC C

CLR D Q +

load_cnt0_i

RTL_ADD

I1

I0[15:0] O[15:0]

load_mode_reg

RTL_REG_ASYNC C

CLR D Q

n_state_i

RTL_MUX S

V=B"001", S=1'b1 I0[2:0] V=B"100", S=default I1[2:0] O[2:0]

n_state_i__0

RTL_MUX S

V=B"001", S=1'b1 I0[2:0] S=default I1[2:0] O[2:0]

n_state_i__1

RTL_MUX

S=16'b0000000011011010 I0[2:0] S=default I1[2:0] O[2:0] S[15:0]

n_state_i__2

RTL_MUX

S=3'b000 I0[1:0] S=3'b001 I1[2:0] S=3'b010 I2[2:0] S=3'b011 I3[2:0] S=3'b100 I4[2:0] S=default I5[2:0]

O[2:0]

S[2:0]

n_state_i__3

RTL_MUX

V=B"0001", S=3'b001 I0[3:0] V=B"0010", S=3'b010 I1[3:0] V=B"0100", S=3'b011 I2[3:0] V=B"1000", S=3'b100 I3[3:0] O[3:0]

S[2:0]

n_state0_i

RTL_MUX S

V=B"01", S=1'b1 I0[1:0] S=default I1[1:0] O[1:0]

n_state0_i__0

RTL_ROM

A[15:0] O[2:0]

n_state0_i__1

RTL_ROM

A[15:0] O[2:0]

!=

n_state1_i

RTL_NEQ O

I0[15:0] V=B"0000000000000111" I1[15:0]

!=

n_state1_i__0

RTL_NEQ O

I0[15:0] V=B"0000000000101111" I1[15:0]

valid0_i RTL_AND

I0

I1 O

valid0_i__0 RTL_AND

I0

I1 O

valid0_i__1 RTL_AND

I0

I1 O

valid0_i__2 RTL_AND

I0

I1 O

valid0_i__3 RTL_AND

I0

I1 O

valid0_i__4 RTL_AND

I0

I1 O

valid0_i__5 RTL_AND

I0

I1 O

valid0_i__6 RTL_AND

I0

I1 O

valid1_i RTL_AND

I0

I1 O <

valid1_i__0

RTL_LT O

I0[15:0] V=B"0000000011010001" I1[15:0]

valid1_i__1 RTL_AND

I0

I1 O <

valid1_i__2

RTL_LT O

I0[15:0] V=B"0000000011010010" I1[15:0]

valid1_i__3 RTL_AND

I0

I1 O <

valid1_i__4

RTL_LT O

I0[15:0] V=B"0000000011010011" I1[15:0]

valid1_i__5 RTL_AND

I0

I1 O <

valid1_i__6

RTL_LT O

I0[15:0] V=B"0000000011010100" I1[15:0]

valid1_i__7 RTL_AND

I0

I1 O <

valid1_i__8

RTL_LT O

I0[15:0] V=B"0000000011010101" I1[15:0]

valid1_i__9 RTL_AND

I0

I1 O <

valid1_i__10

RTL_LT O

I0[15:0] V=B"0000000011010110" I1[15:0]

valid1_i__11 RTL_AND

I0

I1 O <

valid1_i__12

RTL_LT O

I0[15:0] V=B"0000000011010111" I1[15:0]

valid1_i__13 RTL_AND

I0

I1 O <

valid1_i__14

RTL_LT O

I0[15:0] V=B"0000000011011000" I1[15:0]

>=

valid2_i

RTL_GEQ O

I0[15:0] V=B"0000000000001101" I1[15:0]

>=

valid2_i__0

RTL_GEQ O

I0[15:0] V=B"0000000000001110" I1[15:0]

>=

valid2_i__1

RTL_GEQ O

I0[15:0] V=B"0000000000001111" I1[15:0]

>=

valid2_i__2

RTL_GEQ O

I0[15:0] V=B"0000000000010000" I1[15:0]

>=

valid2_i__3

RTL_GEQ O

I0[15:0] V=B"0000000000010001" I1[15:0]

>=

valid2_i__4

RTL_GEQ O

I0[15:0] V=B"0000000000010010" I1[15:0]

>=

valid2_i__5

RTL_GEQ O

I0[15:0] V=B"0000000000010011" I1[15:0]

>=

valid2_i__6

RTL_GEQ O

I0[15:0] V=B"0000000000010100" I1[15:0]